

Abstract

The prediction of chemical reaction properties, like activation energies or rates, has countless applications in developing novel drugs or optimizing industrial processes. Graph neural networks (GNNs) have become a popular machine learning approach to learning molecular representations by aggregating local atom and bond information. Contrastingly, recent work has explored reframing reaction prediction into natural language translation with SMILES representation using the powerful transformer architecture. However, SMILES representations of reactions are blind to atom and bond features and are not permutation invariant. Here, we propose the Spatial Reaction Neural Network (SpatialRXNN), which synthesises previous GNN and transformer approaches into a single architecture that can learn both local chemical features and global interdependencies in chemical reactions. Crucially, SpatialRXNN replaces the positional encodings in a traditional transformer with 3D spatial encodings of the reaction, enabling the model to learn spatial relationships in the reaction that may not be captured by a molecular graph or SMILES string. Finally, by analysing the attention weights in the transformer, SpatialRXNN is interpretable and can generate chemical insights, giving it greater real-world applicability. We show that SpatialRXNN outperforms previous state-of-the-art approaches in three reaction property prediction datasets, highlighting the importance of spatial information in reaction property prediction.